Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

Where's My Bus: A Hardware Independent Bus Tracker

Team Name: *Team #DSCPP-III-2026-T007*	*Null Pointer*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Dangwal, Devansh – 2510370109* *devanshdangwal7@gmail.com* 
Team member 2 *(Last Name, name: student ID: email, picture):*	*Gusain, Aryan – 2510012634* *aryangusain005@gmail.com* 
Team member 3 *(Last Name, name: student ID: email, picture):*	*Lakhera, Vinayak – 2510010715* *vinayaklakhera2@gmail.com* 
Team member 4 *(Last Name, name: student ID: email, picture):*	*Singh, Gautam – 2510012593* *grandmastergm765@gmail.com* 

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

Where is my bus? is the bus monitoring system for a college with pre-determined bus paths. At present, a student does not have an easy mechanism to figure out whether the bus is late, is full, or is out of service. Our application gives every individual a color for each bus at his/her preferred stop point.

There are three types of users. The student user just views the information without logging in to the system. The conductor logs in and updates the status of his/her bus by pressing the button once he/she reaches the next stop point. The Head of Department (HOD) gets the summary report of all the buses.

Colors are determined at the time and not saved: green is for seats empty, blue is for bus full, red is for bus has gone past this stop, brown is for resting, purple is for bus maintenance, and black is for bus not available. All the stops, routes, buses and estimated arrival times are saved in the MySQL database. For now, The entire project is hosted by just one computer without any internet connectivity. We have used handwritten data structures such as linked list, queue, stack, binary heap, and hash table.
We have made this project to be hardware independent and cost effective. No use of hardware (like gps) or costly api would be integrated in future devlopment.

Updated Project Approach and Architecture

The architecture of the project is divided into layers and each layer communicates only with the next one.

1. Database: MySQL is used to store information about stops, routes, their order, buses and their ETAs at certain stops. There are no hard-coded values like names of stops and bus numbers in C++ code, all of this information is taken from database at start.

2. Classes and data structures: each route is represented as a linked list of stops. Routes and buses are stored in hash tables to make their search possible. Conductor's update of bus' position becomes a new element of the queue, then is checked and added to the stack. Binary heap sorts buses for display purposes. Users (student, conductor, HOD) are created via inheritance.

3. Status engine: This module allows to move bus along the route and determine color of the bus at the stop.
4. Display: the C++ program generates the student page and HOD page as HTML pages. Then, the browser loads these pages every two seconds, and thus the colors get updated right after the conductor changes the color on the bus.

Communication between components : The C++ program executes the command line tool of MySQL in the background process, asks a question to it and waits for the printed response.

Technology used: C++(with stl), MySQL 8, g++.

Tasks Completed

Task Performed	Name
Database with 25 stops, 3 routes, 14 buses and ETAs between stops	Vinayak Lakhera
Connection with C++ and MySQL, and import of data in the program	Vinayak Lakhera
Stop, Route (linked list), Bus and Update class	Devansh Dangwal
Bus status engine with moving of bus, colors, completion of trip, undo and redo functions	Devansh Dangwal
User, Student, Conductor, HOD classes; Hash table to search bus, routes; route validation	Vinayak Lakhera
Menu for the conductor and HOD terminal, configuration and main program	Aryan Gusain
Queue, Stack and Binary heap template class and sorting of buses	Gautam Singh
Student and HOD pages created in HTML and CSS	Aryan Gusain, Gautam Singh
Frontend and Backend Connection	Aryan Gusain
Slides for the presentation and Reports	Aryan Gusain, Devansh Dangwal

Challenges/Roadblocks

1. The same stop can appear twice along the way. In Graphic Era stop is both the first stop and the last stop; therefore, we cannot determine the position based on the stop name. Instead we used the stop order number.
2. The view for ETAs in the database showed an incorrect value for the last record in every route. We discontinued the usage of the view and wrote our own query to obtain the correct ETA.
3. Linking of C++ to MySQL on Windows was challenging because of the inability to make connection through the usual libraries. This problem was solved by executing the MySQL command line client application from C++. The issue with paths containing spaces was solved by adding extra quotes to the command.
4. The web browser could open a page even when it was not yet written. Now we write the file to a temporary place and then move the file.
5. Pointers and memory-related issues. We gave every object one unique owner and did tests with memory checkers.
6. Undo was tricky because all buses share one history stack, We take out the newer updates of other buses, undo ours, and put the others back in the same order.
7. Pickup tracking is still not working. Our route linked list only goes forward, so the bus cannot come back from the rest stop to Graphic Era. We plan to use a doubly linked list for this.

Tasks Pending

Pending Task	Team Member (who is going to complete the task)
Conductor Dashboard (web page rather than the terminal menu)	Gautam Singh and Aryan Gusain
Larger Database with greater number of routes, bus stops and buses	Vinayak Lakhera
Pickup Bus Tracking (bus departs from the rest stop and goes backward along the route towards Graphic Era using doubly linked list)	Devansh Dangwal, Vinayak Lakhera
ETA calculation	Devansh Dangwal
More test cases (Missed bus, resting bus and wrong updates)	Gautam Singh, Aryan Gusain
Testing on the laptop which is going to be used for demonstration	All Members
Code review of all the files	All Members
Preparation of User manual with screenshots	Aryan Gusain
Preparation of Final .Report	Devansh Dangwal, All Members
Final Presentation	Aryan Gusain, All Members

Project Outcome/Deliverables

(Key outcomes / Deliverables of the project).

Backend built using C++ programming language for loading routes, buses and their ETAs from MySQL database.
Page for HOD and for students which displays color of each bus for selected bus stop and auto refreshes itself.
Conductor's menu to start a new shift, note down next stop arrived, make a mark of seats available, maintenance and undo.
Report for department head containing bus numbers and updates made recently.
Implementation of linked list, queue, stack, binary heap and hash table.
Source code, SQL files, user manual, final report and demo.

Progress Overview

(How much of the project is done, what's behind schedule, what's ahead of schedule)

Backend is done. Currently, the bus tracking system operates only for the dropping trips, where the bus leaves from Graphic Era and drops the students at the stops.
Tracking system for picking up is yet to be done.
The database is done, it consists of just 3 routes and 14 buses, however there it can be extended.
Student pages and HOD page have been done.
The conductor is using the terminal-based menu and the dashboard is yet to be done.

Late: Pickup tracking, ETA, dashboard for conductor, user manual and final report.
On schedule: Live demo for dropping trip is already functional.

Codebase Information

(Repository link, branch, and information about important commits.)

Repository link: [https://github.com/Nickyyayy/WheresMyBus]
Branch: main (each member also has their own branch)
Important commits: e9452e9 (HEAD -> main, origin/main) -> registries
4250780 -> data structures
c012719 (origin/master) -> Project folder structure,windows build file,load and config file for sql database, core classes implementation
a3cee68 -> Added mysql database + Modified Readme

Folders: backend (all C++ code), frontend (style.css), sql (database files), output (pages made by the program).
The database password is kept in a separate config file which is not uploaded to GitHub.

Testing and Validation Status

(Tests conducted)

Test Type	Status (Pass/Fail)	Notes
Checks done and passed	No warning.	
Memory checks	Pass	No memory leak.
Loading of data from MySQL	Pass	3 routes and 14 buses were successfully loaded.
Conductor's operations (Drop Trip)	Pass	Starting a shift, reaching to stop, seat full, maintenance, undo and trip completion were performed successfully.
Student Page color test	Pass	The colors were matching the expected values.
HOD Report	Pass	Shows the successful trip.
Data structures test	Pass	Heap Order check, Linked List Check and Undo works.
Error Messages	Pass	Incorrect password, missing configuration and incorrect database messages shown.
Pickup tracking test	Fail	Bus is unable to return from rest stop to Graphic Era.

Deliverables Progress

Back end – Done
Student page – Done (drop trip only)
Report generation by HOD – Done
Bus tracking for drop trip – Done
Database – Partly done (extra information can be added)
Conductor menu in terminal – Done
Conductor dashboard – To do
Pick up bus tracking – To do
ETA calculation – To do
Testing – Ongoing
Manual – To do
Final report – To do
Final presentation – To do